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**DASCI Advanced Machine Learning**
FISE A2

# Online competition : Stock Return Prediction by QRT

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# Sommaire

# 1. Context and Motivation

This project was conducted in the context of an online machine learning competition focused on **market-neutral alpha prediction**. The objective was to predict the relative performance of financial assets while remaining neutral to overall market movements. Unlike traditional return prediction problems, this setting imposes strong constraints : signals must capture *cross-sectional information* rather than market trends, and performance is evaluated in a way that penalizes exposure to systematic risk.

This challenge was particularly useful from a learning perspective because it sits at the intersection of **financial modeling**, **time-series feature engineering**, and **robust machine learning evaluation**. It required careful reasoning about data leakage, temporal structure, and the limits of classical metrics when applied to noisy financial data.

# 2. Data Understanding and Exploratory Analysis

The dataset consisted of lagged financial features, mainly returns and volumes, provided over rolling time windows. A first critical difficulty was that the data was already highly engineered and anonymized, which prevented the use of domain-specific indicators in their usual form. As a result, the exploratory data analysis (EDA) focused less on semantic interpretation and more on **statistical behavior**.

Key observations from the EDA included :

- The target variable was almost perfectly balanced, ruling out trivial baselines.
- Individual lagged returns showed very weak marginal predictive power.
- Volume-related features exhibited regime-like behavior, suggesting non-linear interactions.
- Strong noise dominance and low signal-to-noise ratio, typical of financial prediction tasks.

This analysis confirmed that simple linear models were unlikely to perform well and motivated the use of non-linear ensemble methods.

# 3. Baseline Model and Evaluation Strategy

As a first reference point, we implemented a **baseline model** using a limited set of lagged returns and volumes. The goal was not performance optimization but rather :

- To validate the full pipeline (data loading, splitting, evaluation),
- To obtain a realistic lower bound on achievable performance.

The baseline achieved only a marginal improvement over random guessing, which was expected given the difficulty of the task. However, this step was essential to :

1. Detect potential data leakage,
2. Validate temporal consistency in the train/validation split,
3. Provide a reference for measuring real progress.

# 4. Feature Engineering

The most impactful part of the project was feature engineering. Since raw features carried little standalone information, we focused on extracting **aggregated and regime-aware signals**. The main strategies included :

- Rolling statistics (means and volatilities) over multiple horizons,
- Cross-feature interactions between returns and volumes,
- Discretization of volatility regimes to capture non-linear market states.

These transformations significantly improved the expressive power of the feature set while remaining compatible with tree-based models. Importantly, all features were constructed using strictly past information to avoid look-ahead bias.

# 5. Final Model and Rationale

The final modeling approach relied on an ensemble of non-linear tree-based models, including Random Forests, XGBoost, and LightGBM. This design choice was motivated by the highly noisy nature of financial data and the weak predictive power of individual signals.

Tree-based ensembles are particularly well-suited to this setting as they can capture complex non-linear interactions, regime-dependent effects, and heterogeneous feature contributions without requiring strong parametric assumptions. Gradient boosting models such as XGBoost and LightGBM further provide robustness through regularization mechanisms including subsampling, shrinkage, and depth constraints.

Rather than relying on aggressive hyperparameter optimization, the emphasis was placed on feature engineering quality and validation discipline. Model selection was performed using date-based cross-validation to limit cross-sectional leakage, and the final predictions were obtained by averaging the probabilistic outputs of the different models, leading to a more stable and robust final submission.

# 6. Challenges Encountered

Several challenges were encountered throughout the project :

- **Extreme noise** : most features had very weak predictive power.
- **Evaluation ambiguity** : small metric improvements are hard to interpret statistically.
- **Data management** : large files required careful repository organization and exclusion from version control.
- **Collaboration constraints** : merging parallel work streams required strict structure and reproducibility.

Addressing these challenges required both technical rigor and practical engineering decisions.

# 7. Skills and Techniques Acquired

This competition allowed us to develop several advanced skills :

- Designing leakage-free pipelines for time-dependent data,
- Interpreting weak signals in high-noise environments,
- Structuring a professional machine learning repository,
- Collaborating through Git in a multi-branch workflow.

Beyond the final score, the main value of this project lies in the methodological discipline it enforced, which closely mirrors real-world quantitative research constraints.

# 8. Conclusion

In conclusion, this challenge provided a realistic and demanding setting to apply advanced machine learning techniques to financial data. The final solution succeeded not because of excessive model complexity, but due to careful feature engineering, disciplined evaluation, and a clear understanding of the problem structure. These lessons are directly transferable to real-world applied machine learning and quantitative finance problems.